

Architecture Decision Record (ADR): Model Selection for Trip Duration Prediction

1. Which Model?

Decision: We have chosen the **Gradient Boosting Regressor (GBR)** as the primary model for predicting trip duration.

2. Why this Model?

The Gradient Boosting Regressor was selected for the following reasons:

- **Slightly Superior Performance:** When comparing the initial training runs, the Gradient Boosting Regressor exhibited a marginally higher R-squared value (0.0290) compared to the Linear Regression model (0.0279). While both values are low, indicating limited predictive power with the current feature set, GBR still provided a fractional improvement in explaining the variance in trip duration.
- **Robustness:** Gradient Boosting models are generally more robust to overfitting than single decision trees and can capture non-linear relationships within the data, which Linear Regression cannot. This makes it a more versatile choice for complex, real-world datasets.
- **MLflow Integration:** Both models were tracked using MLflow, which allowed for systematic logging of parameters, metrics (MSE, R-squared), and model artifacts. This facilitates easy comparison and versioning of model runs.

3. What Alternatives Were Considered?

Alternative: Linear Regression was the primary alternative model considered.

- **Evaluation:** Linear Regression was trained and evaluated alongside the Gradient Boosting Regressor. Its R-squared score was slightly lower (0.0279) than that of the GBR (0.0290), and its Mean Squared Error (9,516,939.78) was marginally higher than GBR's (9,505,976.93).
- **Reason for not choosing:** While Linear Regression is simpler and more interpretable, its performance was slightly inferior for this specific problem according to the chosen evaluation metric (R-squared). Given the objective of finding the 'better' performing model among the two, GBR was preferred.

4. What Trade-offs Were Accepted?

Several trade-offs were accepted in this model selection and development process:

- **Low Predictive Performance:** Both models (Linear Regression and Gradient Boosting Regressor) achieved very low R-squared scores (approximately 0.03). This indicates that the models, even the Gradient Boosting Regressor, explain only a small fraction of the variance in trip duration. This suggests that the problem of predicting trip duration with the current feature set is inherently complex, or critical features influencing trip duration (e.g., real-time traffic data, driver behavior, event data) are missing from the dataset. We accepted this trade-off for the scope of this mini-project, acknowledging that significant model improvements would likely require more advanced feature engineering, additional data sources, or more sophisticated models (e.g., deep learning).
- **Computational Complexity vs. Simplicity:** Gradient Boosting Regressor is computationally more intensive during training than Linear Regression. While GBR offered slightly better performance, this came with an increased computational cost. For this project, the performance gain was deemed worthwhile for the marginal increase in complexity.
- **Interpretability vs. Performance:** Linear Regression offers high interpretability, with clear coefficients indicating feature impact. Gradient Boosting models, while more powerful, are less interpretable (often considered 'black boxes'). We prioritized the slight performance improvement of GBR over the higher interpretability of Linear Regression.
- **Limited Hyperparameter Tuning:** While hyperparameter tuning for the Gradient Boosting Regressor was initiated using GridSearchCV (as shown in cell 8e29a76), it was not fully executed within the scope of this mini-project. This means the chosen GBR model's performance might not be fully optimized. This trade-off was made to maintain focus on the MLflow integration and data drift monitoring aspects of the project, acknowledging that further performance gains could be achieved with more extensive tuning.
- **Data Drift Sensitivity:** The model's performance significantly degraded on simulated drifted data (R-squared dropped from 0.0279 to 0.0170 for Linear Regression, triggering retraining). This highlights that the model is sensitive to changes in the data distribution. We accepted the necessity of implementing continuous monitoring and retraining mechanisms as an operational trade-off, rather than seeking a single, universally robust model that might not exist given the dynamic nature of the data.